IMEN891K / IDSC501 / AXGS-501 · DATA ENGINEERING · FALL 2026

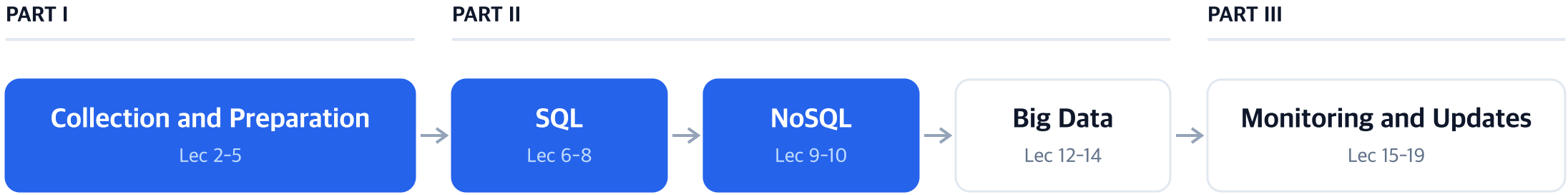
11.

Review Session

Sangwoo Mo · POSTECH

Industrial & Management Engineering

Where We Are


PART I
PART II
PART III
Collection and Preparation
Lec 2-5
SQL
Lec 6-8
NoSQL
Lec 9-10
Big Data
Lec 12-14
Monitoring and Updates
Lec 15-19

Outline: Review Session

01	**Collection and Exploration**	Lec 2-3 · records, labels and patterns
02	**Data Preparation**	Lec 4-5 · changes to values and representation
03	**SQL**	Lec 6-8 · read, match and summarize rows
04	**NoSQL**	Lec 9-10 · models, documents and updates

01 · Collection and Exploration

01 **Collection and Exploration** — Lec 2-3 · records, labels and patterns

02 Data Preparation — Lec 4-5 · changes to values and representation

03 SQL — Lec 6-8 · read, match and summarize rows

04 NoSQL — Lec 9-10 · models, documents and updates

Outline: Data Collection



Online and Offline Data

Online: digital assets and logs

Where the record is exposed	How to obtain it
A system provides an interface	API calls
Only a webpage is available	Web crawling

Offline: physical observations

Measure a physical signal with a sensor, or record a human observation.

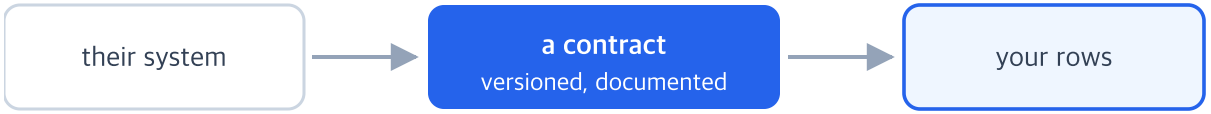
The measurement must become a digital record.

Locate the observation source before choosing the collection method

API or Crawler

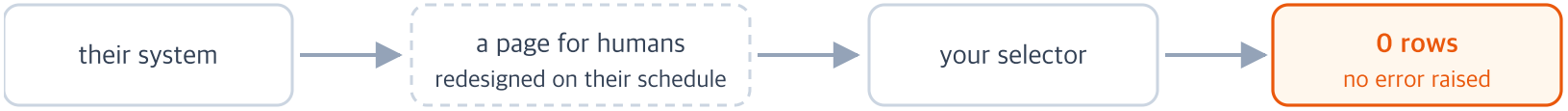
Prefer an explicit interface; page extraction also depends on the page layout

API



An explicit interface can be versioned and checked

CRAWLER



A redesign breaks extraction without an HTTP error

An explicit interface can be checked; page extraction also depends on layout

Sensors as Encoders

Physical signal	Encoder	Recorded representation
Temperature	Temperature sensor	A timestamped temperature value
Vibration	Vibration sensor	A sequence of sampled values

The sensor converts a physical signal into a form the system can record digitally.

Measurement turns a physical event into a recorded observation

Record Meaning and Context

A recorded value needs	Example
An entity and event time	Part P-104 at 09:14:02
A measurement and unit	Cycle time: 38.6 seconds
A source and definition	CNC-02 log; defect rule v3

A value becomes usable when its meaning and context are recorded

Generating Training Examples

Approach	What changes	What needs checking
Annotation	Add a label to an existing input	Definition and label quality
Augmentation	Change an input while keeping its label	Whether the label still applies
Synthesis	Generate a new input	Realism, coverage and usefulness

Generated inputs and candidate labels still require validation

Sources of Labels

A label can come from deliberate annotation or from an observed outcome

Source	Example	Main collection problem
Human annotation	An inspector judges a part	Make the guideline reproducible
Natural label	A delivered shipment has an arrival time	Record and link the outcome

Model-generated labels remain candidates until checked

Both sources need a definition and a quality check

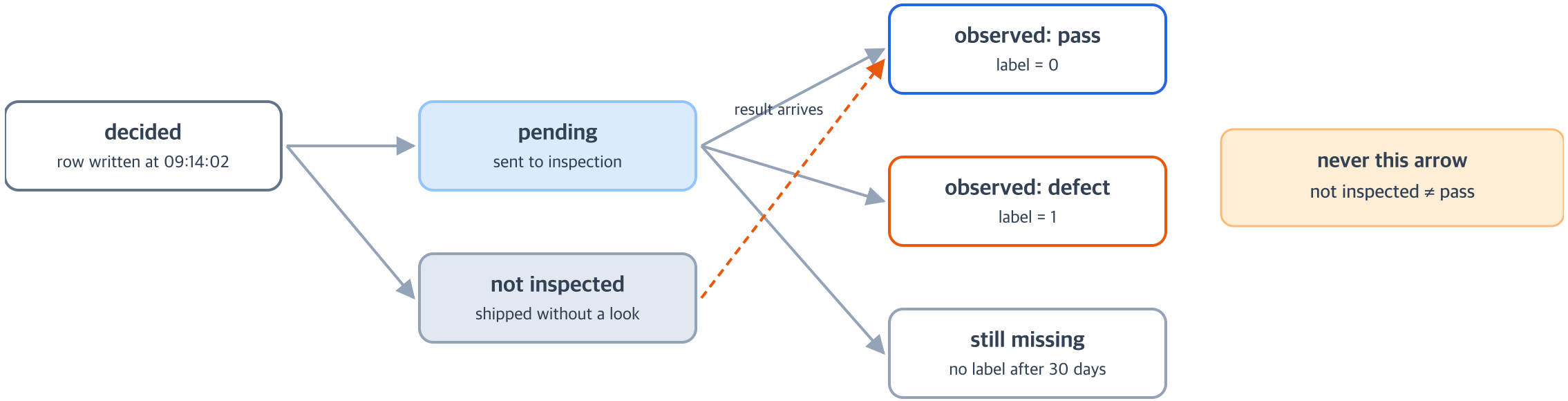
Checking Label Quality

Check	Question
Definition	Does defect mean a tolerance violation, a return, or a complaint?
Agreement	Do reviewers applying the same guideline agree?
Independent audit	What outcomes does the current selection policy miss?

A label needs a clear definition, reproducible judgment and independent checks

Missing Outcomes

Keep pending, unobserved and negative outcomes distinct



A model trained on "label or 0" learns that skipped parts are good

Not inspected is different from an observed pass

Selected Outcomes

The inspection policy controls which outcomes become labels

100 produced parts

20 selected for inspection

Labels will be observed

→

80 shipped without inspection

Outcomes unknown

They are not recorded as pass

→

Random audit of 5

25 labels in total

75 outcomes remain unobserved

Audit outside the model-selected sample to observe what it missed

Collection Boundaries

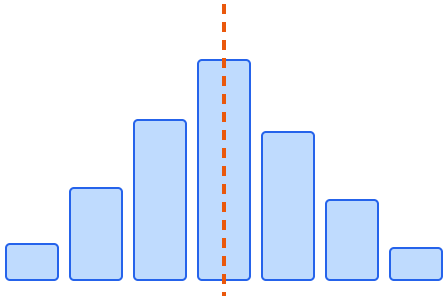
Before collecting or reusing	Record the decision
Purpose and access	Why the data is needed and who may use it
Rights and confidentiality	Whether collection, reuse and transfer are permitted
Protection and provenance	Restrict access; retain source, time and definition

Collect only what the purpose needs, with permission and protection

Outline: Data Exploration

01

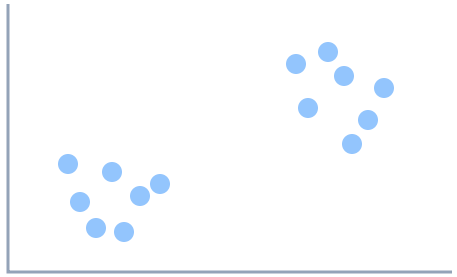
Descriptive Analysis



Reduce the data to a few numbers

02

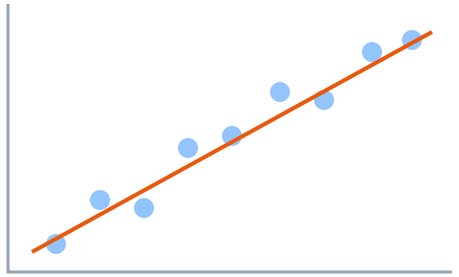
Visual Exploration



Find patterns by plotting the data

03

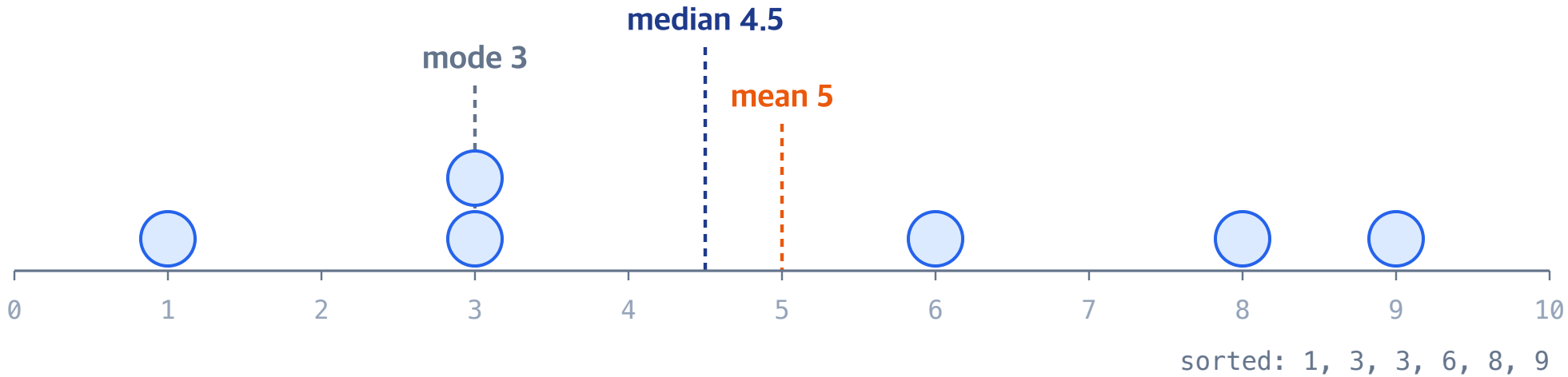
Statistical Exploration



Read relationships between variables

Recap: Mean, Median, Mode

Three answers to “the typical value”



From 9, 3, 1, 8, 3, 6: mean **5**, median **4.5**, mode **3**

Extreme Values and the Median

Sorted values	Mean	Median
1, 3, 3, 6, 8, 9	5	4.5
1, 3, 3, 6, 8, 99	20	4.5

The mean uses every value; the median uses the middle positions.

An extreme value pulls the mean; the median is less sensitive

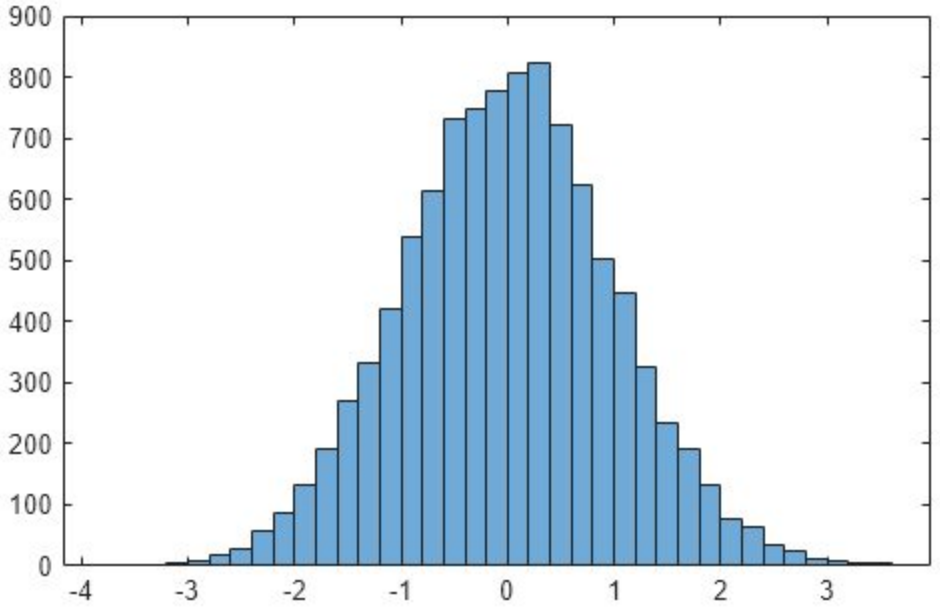
Variability and Robust Summaries

Summary	What it describes	Effect of extreme values
Mean and standard deviation	Center and spread using all values	Can change substantially
Median and interquartile range	Middle value and middle half of the data	Less sensitive to extreme values

Report spread as well as center, and inspect the tails

Histogram as Empirical Distribution

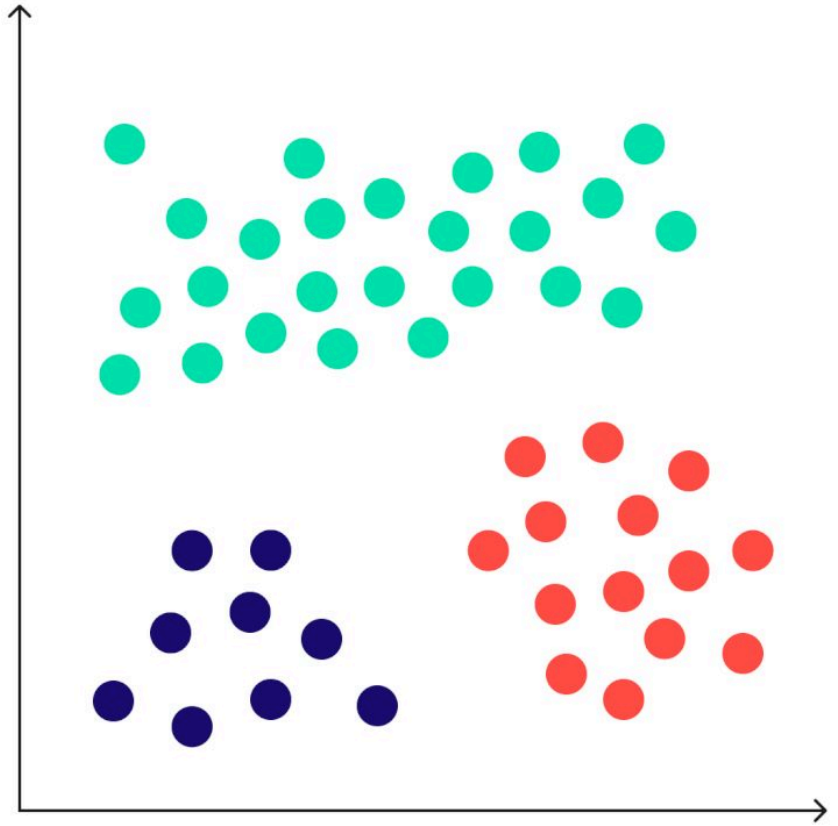
Bins summarize how many observations fall in each interval



Bin counts show distribution shape while hiding within-bin detail

Clustering with k-means

It always returns groups: the same points after asking k-means for three



Choosing k asks for that many clusters; it does not establish the true group count

PCA and t-SNE

Method	What to look for	What the plot does not establish
PCA	Directions retaining large variance	The most predictive or causal directions
t-SNE	Local neighborhoods in a low-dimensional view	Reliable distances between separate clusters

A projection is a view of the data, with method-specific limits

Time Series Views

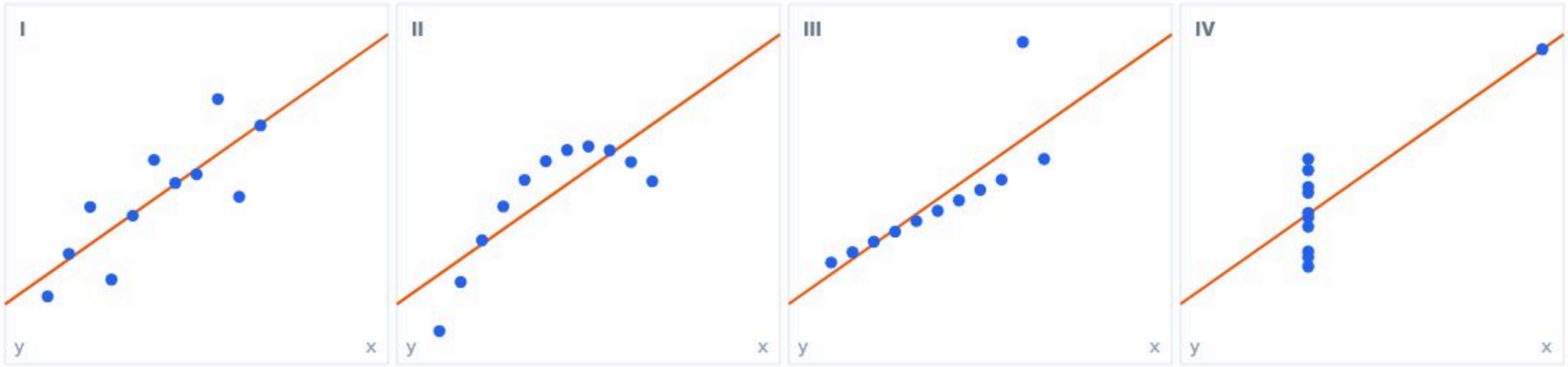
View	Question
Time domain	When did the level, trend or variation change?
Frequency domain	Which repeating cycles contribute to the signal?
Across time periods	Is the process stable enough for the comparison?

Time plots compare values over time; scatter plots compare two variables.

Preserve time order and check whether the process changes

Anscombe’s Quartet

Similar summary statistics, different scatter plots



The same summaries can hide different relationships

Pearson and Spearman Correlation

Measure	What it summarizes	Main limit
Pearson	Strength and direction of a linear association	A nonlinear relation can have a small coefficient
Spearman	Association between the ranks of two variables	Rank association does not imply causation

Choose a relationship summary, then check the scatter plot

Causal Graphs

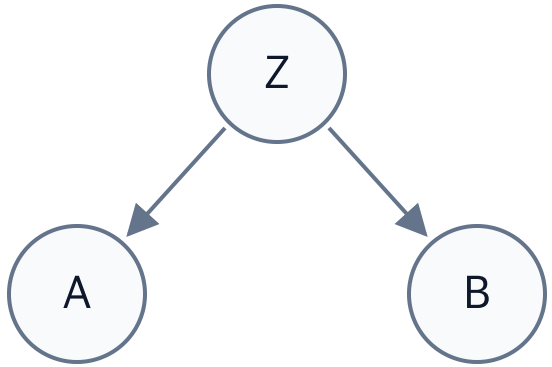
The same correlation can reflect either causal direction or a shared cause



A causes B



B causes A



Z causes both

Three structures give one correlation; **the correlation alone cannot tell them apart**

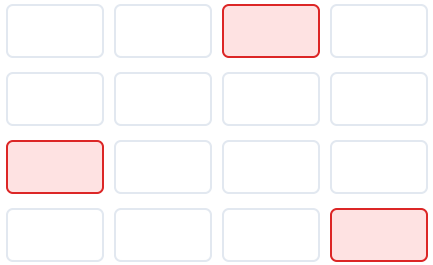
02 · Data Preparation

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Outline: Four Stages of Preprocessing

01

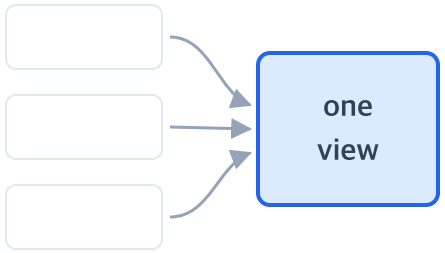
Cleaning



Wrong values in one table

02

Integration



Tables that disagree

03

Transformation



Values a model can read

04

Reduction



Less data, same signal

Data Imputation

Imputation estimates a missing value using observed data.

Possible response	When deciding, check
Keep missing or remove a row	Whether absence has meaning and which records would be lost
Fill from a column summary	Whether a typical value is a useful estimate
Predict from related observations	Whether the other values carry relevant information

An imputed value is an estimate, not a new observation

Observed and Filled Values

Use other observed columns to find comparable rows

Observed measurements

part	helper	target
P1	10	20
P2	12	24
P3	11	NULL

Estimate for P3

Neighbors in helper: P1 and P2

(20 + 24) / 2 = 22

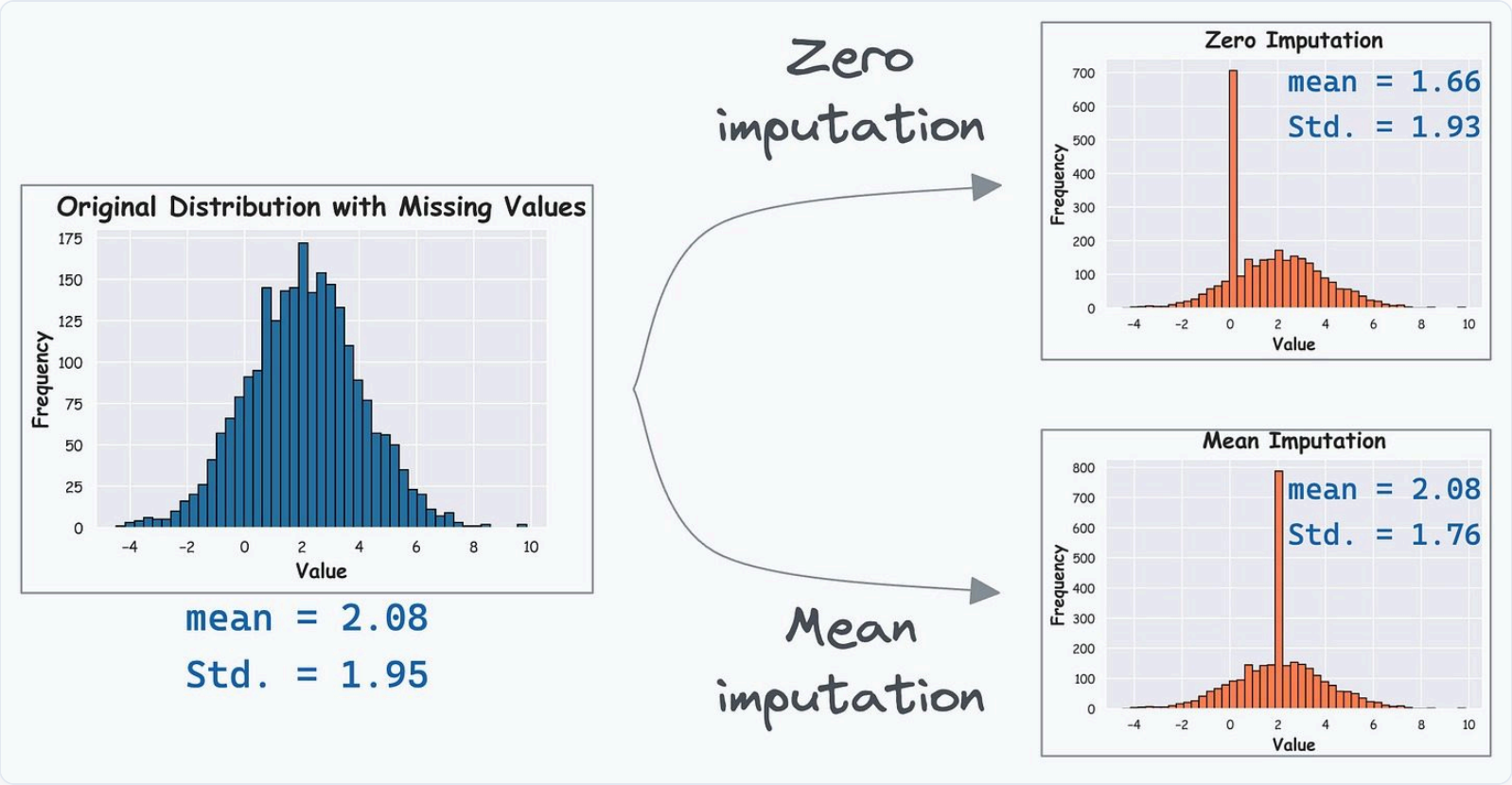
Stored with provenance

part	target	estimated
P3	22	Yes

The filled number is an estimate; keep its origin distinguishable

Caveat: Mean Imputation

Every hole gets the same value, and the spread of the column collapses



Mean stays 2.08; standard deviation falls to 1.76 and a spike appears

Caveat: Why is it missing?

Holes independent of the value

age	25	26	29	30	30	31	44	46	48	51	52	54
IQ	×	121	91	×	110	×	118	93	×	×	116	×

Holes concentrated in the younger half

age	25	26	29	30	30	31	44	46	48	51	52	54
IQ	×	×	×	×	×	×	118	93	116	141	97	104

Below, no young IQ was ever observed. **No imputer fixes this.** The question is not which method to use but why those rows are missing, and that answer lives in how the data was collected

Noise vs Outliers

Smooth measurement noise; investigate outliers before changing them

Noise: every value is a little wrong

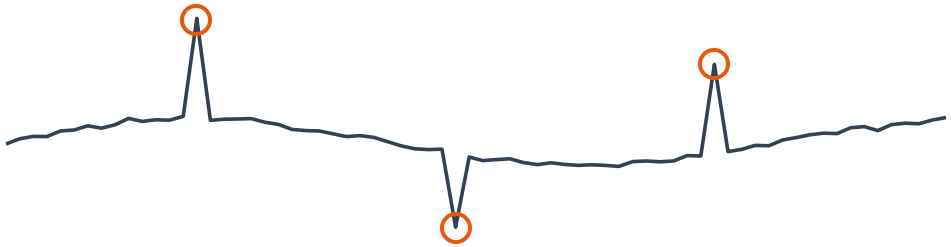


smooth

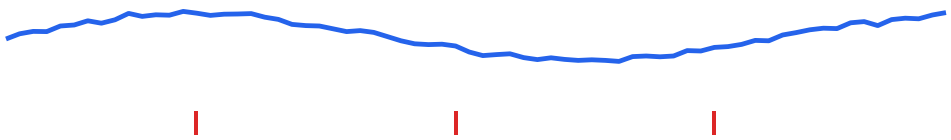


all 1,200 rows survive

Outliers: a few values sit far away



if confirmed: remove

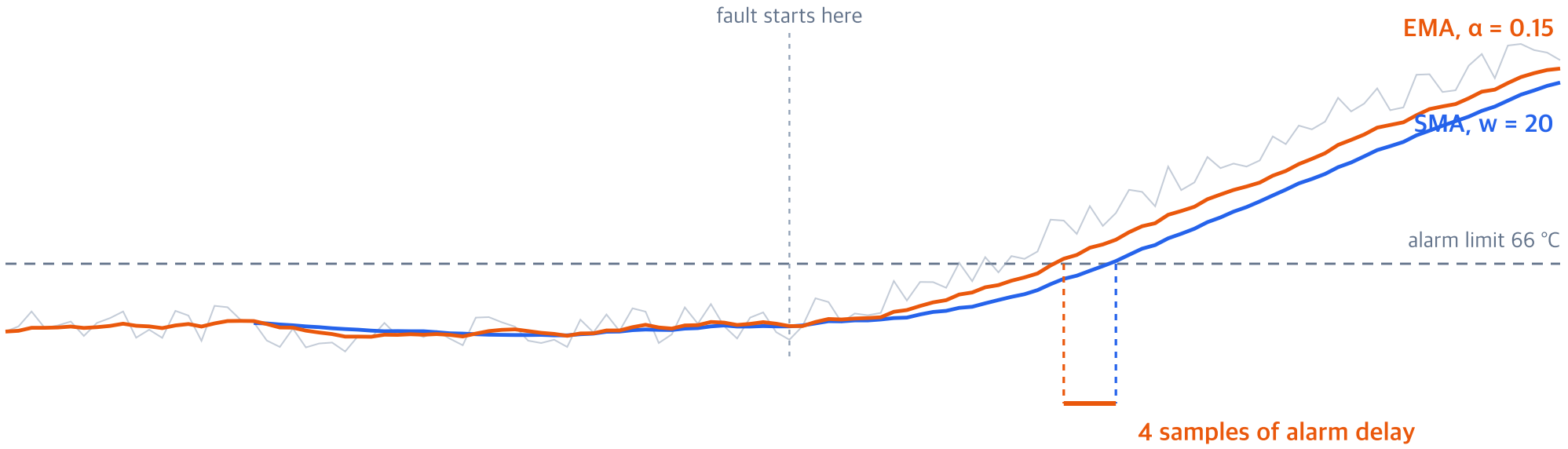


three rows are gone

Smoothing changes values; removal drops rows

Moving Averages for Streams

Simple (SMA) and exponential (EMA) moving averages use only current and past samples



Both use past samples; here EMA crosses four samples before SMA

Consistency Checks

Rule	What a violation tells us
A time interval cannot end before it starts	The record needs investigation
A value must use the declared unit and type	The representation conflicts with its definition
A reference must identify an existing entity	The relationship is incomplete or wrong

A failed check identifies a problem, not its correct replacement

One Customer, Three Systems

System	Source ID	Customer name
CRM	C-10482	Hanil Precision Co., Ltd.
SCM	V0000734	HANIL PRECISION
ERP	2010-HP	한일정밀(주)

Different source IDs and spellings do not establish a shared identity

Three Questions of Integration

Decision	Question	Renaming columns is enough?
Schema mapping	Which columns have the same meaning?	Sometimes, with matching units and definitions
Entity matching	Which rows describe the same customer?	No: different IDs need evidence of identity
Value fusion	Which value should survive a conflict?	No: choose a source or reconciliation rule

Matching columns and matching entities solve different problems

Resolving Conflicting Values

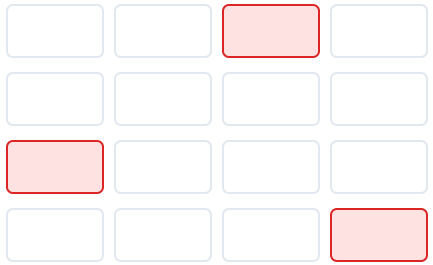
Decision after matching	Example rule
Trust a source for a specific field	Use Finance for recognized revenue
Compare event time and meaning	A newer contact number may replace an older current number
Keep the evidence	Retain source IDs and the reason for the selected value

Entity matching and value reconciliation are separate decisions

Recap: Four Stages of Preprocessing

01

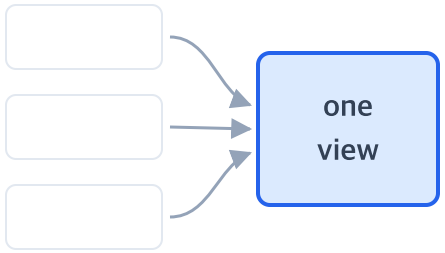
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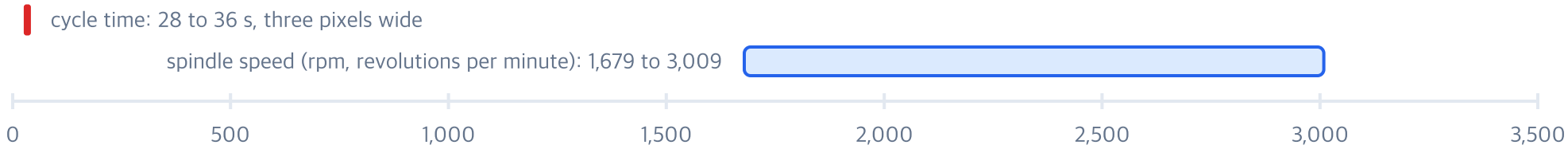


Less data, same signal

Why Scale at All

Distance-based models add up raw units, so the widest column decides

two columns from one machine, on one raw axis



how far apart two rows can be, column by column



In raw Euclidean distance, the rpm difference **dominates** the cycle-time difference

Choosing a Numeric Transformation

Transformation	Purpose	What to check
Min-max or standardization	Put numeric columns on comparable scales	Which training values determine the scale
Robust scaling	Use median and interquartile range	Whether extreme values dominate ordinary estimates
Log transform	Compress a long positive tail	The valid range and changed meaning of differences
Unit-norm normalization	Compare direction rather than magnitude	Whether magnitude carries useful information

Choose a transformation for the comparison the model needs

Encoding, Embeddings and Binning

Operation	Representation change	Information to preserve
Category encoding	A category becomes numeric features	Membership and any intended order
Embedding	An entity becomes a learned vector	What similarity the representation has learned
Binning	A continuous value becomes an interval label	Threshold meaning; exact values are discarded

Encoding changes representation; binning also loses within-bin detail

Nominal and Ordinal Categories

A numeric code can introduce order and spacing into the model

Nominal: no known order

Color	Arbitrary code
red	1
green	2
blue	3

Ordinal: a known order

Size	Ordered code
small	1
medium	2
large	3

Color codes invent an order; size codes express an existing order

One-Hot Encoding

Size	Small	Medium	Large
small	1	0	0
medium	0	1	0
large	0	0	1

These columns identify the category. They do not express small < medium < large.

One-hot represents membership; an ordinal encoding must preserve the intended order

Constructing Features

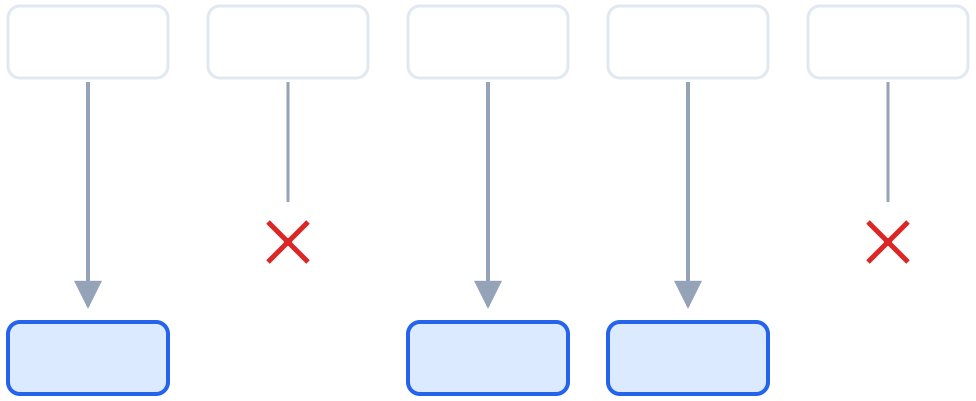
Source of a feature	Example role
Written by a person	Combine measured fields using domain knowledge
Learned from the data	Extract a compact representation
Borrowed from a trained model	Reuse an embedding and check its fit to the task

A useful feature represents information available when the decision is made

Extraction or Selection

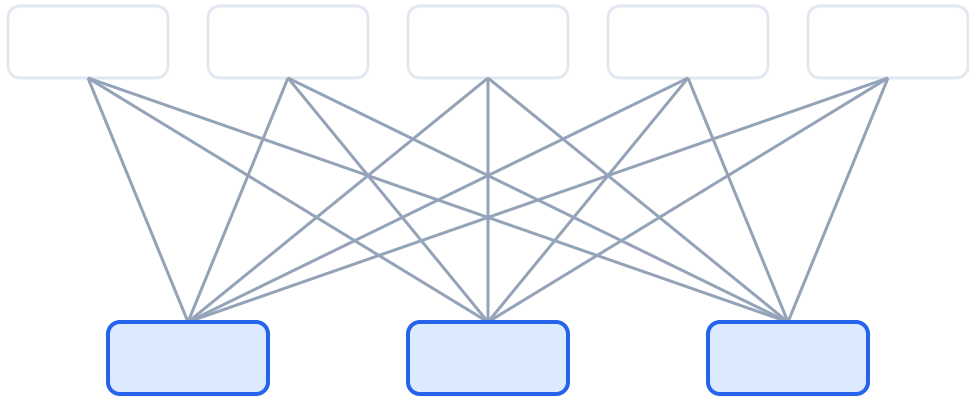
Keep some original columns, or mix all of them into new ones

SELECTION keep some of the original columns



5 columns in, 3 of the same columns out

EXTRACTION make new columns from all of them



5 columns in, 3 new columns out

Selection keeps original columns; extraction creates new combinations

Sampling vs Curation

Decision	Sampling	Curation
Selection criterion	A sampling design	Quality and coverage rules
Main purpose	Use fewer rows for a defined population	Keep useful and reliable examples
What to inspect	Inclusion probabilities and group weights	Which groups and examples were removed

Both select rows; their criteria and resulting biases can differ

Sampling Design

The selection rule determines which parts of the population survive

Method	How rows are chosen	Failure to check
Simple random	Draw rows at random	Small groups may be missed
Systematic	Every kth record	A periodic list can align with k
Stratified	Randomly sample at least one per group	Keep the intended group proportions
Cluster	Select whole groups	One factory may not represent the rest

Check the population structure before choosing the sampling rule

Lossless or Lossy

Compression	After decompression	What to check
Lossless	Exactly the original data	Storage saved with exact recovery
Lossy	An approximation of the original	Whether the discarded information matters

A smaller file is useful only if the retained information still serves the task

The Evaluation Must Match Deployment

Define what will be new at prediction time before splitting the records

Deployment question	Useful split	What a random row split can hide
Predict future events	Earlier train; later test	Future information reaches training
Predict new machines	Hold out whole machines	Same-machine patterns occur on both sides
Predict new customers	Hold out whole customers	Repeated identity makes the test easier

A held-out row can still share an entity or future information with training

Fitting Preprocessing Steps

Step	Learn from training data	Reuse for test and future data
Imputation	A replacement value	Use the same replacement rule
Scaling	Center and spread	Use the same center and spread
Feature selection	Which columns to keep	Keep the same columns

Using test data to choose a transformation leaks information into training.

Fit on training data, then apply the learned transformation consistently

03 · SQL

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Outline: SQL I

01	**Read a Table**	find rows and choose columns
02	**Refine a Result**	remove repeats, sort and keep top rows
03	**Handle Missing Values**	choose how absent prices affect the result
04	**Write Safely**	keep constraints and related changes together

Schema and Rows

A schema defines the columns; rows supply their values

Schema: the definition

```
Product
  pname          TEXT, primary key
  price          REAL
  category       TEXT
  manufacturer   TEXT
```

Rows: the stored values

pname	price	category
Gizmo	19.99	Gadgets
PowerGizmo	29.99	Gadgets
SingleTouch	149.99	Photography
MultiTouch	203.99	Household

One definition, many product rows

Keys and References

cname	stockprice	country
GizmoWorks	25.0	USA
Canon	65.0	Japan
Hitachi	15.0	Japan

pname	manufacturer
Gizmo	GizmoWorks
PowerGizmo	GizmoWorks
SingleTouch	Canon
MultiTouch	Hitachi

A primary key identifies a row; a foreign key requires an existing referenced key

Company facts are stored once; product rows refer to them by key

SELECT, FROM and WHERE

The question names one input table, a row condition and output columns

FROM

Read **Product**

→

WHERE

Keep **Gadgets**

→

SELECT

Show **name and price**

```
SELECT pname, price
FROM Product
WHERE category = 'Gadgets';
```

Read the table, keep matching rows, then choose columns

Refining a Query Result

Decision	Meaning
Repeated rows	Decide whether each occurrence matters or only unique values matter
Ordering	Specify the ranking needed for the question
A limited result	Use a stable order before keeping only the first rows

Rows, distinct values and a ranked list answer different questions

Missing Values in a Table

Stored meaning	Interpretation
A recorded zero	The value is known and equals zero
A missing value	The value is absent, unknown or not applicable
A filter using that field	Decide explicitly how missing records belong in the result

Missing is a state of knowledge, not an ordinary numeric value

Safe Changes to Related Facts

Protection	Reason
Keys and references	Reject duplicate identities and broken relationships
A deliberate target set	Check which rows a change will affect
A transaction	Commit related changes together, or leave all of them unapplied

Preserve constraints and change related facts as one unit

Outline: SQL II

01	**Match Rows**	return one row per matching pair
02	**Keep Unmatched Rows**	keep the left row when no match exists
03	**Test for Matches**	keep each outer row if a match exists
04	**Stack Results**	append compatible result rows

Matching a Product to Its Company

The shared company name lets one result contain product and country

Product

pname	manufacturer
Gizmo	GizmoWorks

→

Match the keys

```
manufacturer = cname
```

→

Company

cname	country
GizmoWorks	USA

Gizmo and USA meet through the same company key

JOIN on Related Keys

```
SELECT P.pname,P.price,C.country
FROM Product P JOIN Company C
  ON P.manufacturer=C.cname;
```

↑ match a product to its maker

pname	price	country
Gizmo	19.99	USA
PowerGizmo	29.99	USA
SingleTouch	149.99	Japan
MultiTouch	203.99	Japan

Each product matches its company; the result has four product rows

Pairs, Required Records and Match Existence

Question	What one output row represents
Which records are related?	One matching pair; multiple matches can repeat an entity
Which starting records must remain?	The required entity, even when related information is absent
Does a related record exist?	One starting entity, regardless of how many matches exist

Choose the output entity before deciding how records are combined

Matching and Stacking Results

Operation	What it does	Required compatibility
Match records	Combine related facts side by side	Keys must refer to the same kind of entity
Stack records	Append rows from multiple results	Columns must have compatible meanings and types

For overlapping lists, decide whether repeated rows should remain.

Matching connects related facts; stacking combines compatible lists

Outline: SQL III

01	**Calculate Measures**	calculate a quantity from detail rows
02	**Summarize Groups**	choose the key and keep one row per group
03	**Reuse Summaries**	use a computed result as another input
04	**Keep Detail Rows**	add group context without merging rows

Aggregate Functions

Function	Question
COUNT(*)	How many purchase rows?
SUM(quantity)	How many units?
SUM(price*quantity)	How much revenue?
AVG(price)	What is the average recorded unit price?
MIN(price), MAX(price)	What are the smallest and largest prices?

Choose the expression that measures the quantity named in the question

Quantities and Weights

Group	Total	Count	Mean
A	10	1	10
B	180	9	20
Combined	190	10	19

(10 + 20) / 2 = 15 treats the groups equally; 190 / 10 = 19 treats observations equally.

The denominator defines which units receive equal weight

Analytical Result Shapes

Detail input

product	price
bagel	1.0
bagel	1.5
banana	0.5
banana	1.0

GROUP BY

product	mean
bagel	1.25
banana	0.75

Grouping turns detail records into one summary row per group

Grouping and Average

One row per product group

```
SELECT product, AVG(price) AS mean
FROM Purchase
GROUP BY product;
```

Result

product	mean
bagel	1.25
banana	0.75

The grouping key defines each result row; AVG defines its measure

A Query Produces a Table

First query: mean by product

product	mean
bagel	1.25
banana	0.75

Next query: keep mean ≥ 1.0

product	mean
bagel	1.25

A query returns a table that another query can use as input

Detail, Summary and Group Context

Output form	Result unit	Rows for the four purchase records
Detail	One purchase	4
Grouped summary	One product group	2
Detail with group context	One purchase plus its product average	4

Group information can summarize rows or accompany the original detail rows

04 · NoSQL

01	Collection and Exploration	Lec 2-3 · records, labels and patterns
02	Data Preparation	Lec 4-5 · changes to values and representation
03	SQL	Lec 6-8 · read, match and summarize rows
04	**NoSQL**	Lec 9-10 · models, documents and updates

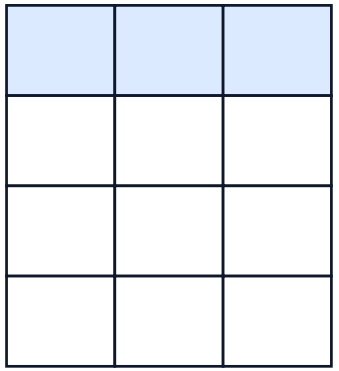
Outline: NoSQL I

01	**Four Families of NoSQL**	choose a model by its access pattern
02	**JSON and Semi-Structured Data**	read named fields and ordered arrays
03	**Between Trees and Tables**	convert nesting and repeated values
04	**Why NoSQL**	check flexibility, scale and guarantees

Four NoSQL Families

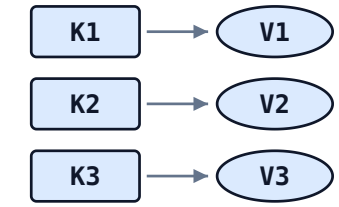
The four NoSQL families

SQL

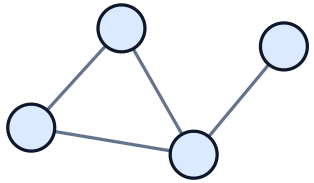


Relational

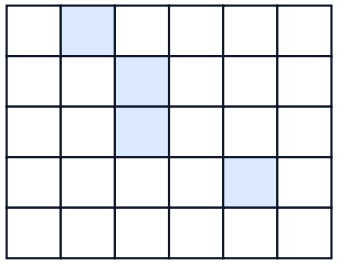
NoSQL



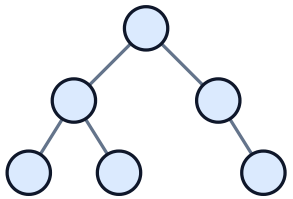
Key-value



Graph



Wide-column



Document

NoSQL is not one thing: four alternative **storage models**

Key-Value Records

Request

```
get("user:1")
```

→

Stored pair

```
key: "user:1"
value: {
  name: "John",
  age: 28
}
```

→

Answer

```
{ name: "John", age:
28 }
```

A key retrieves its associated value

Sparse Cells Under a Row Key

In a Bigtable-style model, rows can hold different column qualifiers

user:1

Family: activity	Value
page:home	09:01
page:product	09:03
purchase:1	Laptop

user:2

Family: activity	Value
page:home	09:02
purchase:2	TV

Same column family, different populated cells

Property Graphs

Graph element	What it records
Node	An entity, such as a person
Edge	A relationship, such as FRIENDS_WITH
Properties	Attributes attached to nodes or edges

To find friends of friends, follow friendship edges from one person to the next.

Graph models make relationships and paths explicit

Evolving Document Shapes

One collection

```
{"name":"John", "age":28}
{"name":"Sue", "phone":"555-0134"}
```

Different optional fields

John has an age field.

Sue has a phone field.

Both documents can belong to the same collection.

Optional fields may differ; shared fields still need consistent meanings

Semi-Structured Data

Relational representation	Document representation
Columns are declared in a schema	Field names appear inside each record
Rows follow the declared table structure	Records may contain nested objects and arrays
Related facts can be linked by keys	Related facts may be nested in one document

Semi-structured data carries structure with the values

Trees and Tables

Representation	Relationship structure	Tradeoff
Nested document	A parent holds related objects or arrays	Read together, but copied facts need coordinated updates
Related tables	Separate rows connect through keys	Facts can be maintained separately; queries reconnect them

Choose nesting for the access pattern and account for repeated facts

Caveat: Stale Copies

Two documents cache the current catalog price, but only one copy receives the update

John's Document

```
{"name": "John",
 "Orders": [
   {"date": 2004,
    "product": {"prodName": "Gadget",
                "currentCatalogPrice": 12.99}}
 ]}
```

Sue's Document

```
{"name": "Sue",
 "Orders": [
   {"date": 2002,
    "product": {"prodName": "Gadget",
                "currentCatalogPrice": 9.99}}
 ]}
```

Remedy: reference the current price or maintain every cached copy

One product, **two prices**: copies need an owner who updates them all

Partitioning and Replication

Splitting records and copying records solve different problems

Partition: different records

```
Server A: users 1–100
Server B: users 101–200
```

Replicate: copies of records

```
Leader: user 1, value 2
Follower: user 1, value 1
          update pending
```

Partitions divide the data; replicas copy it and must coordinate updates

Scaling and Operation Guarantees

Question before choosing a store	What to check
How does capacity grow?	A larger machine, partitions, or both
How do readers see updates?	Whether a read can return a lagging replica
Which changes are atomic?	One record or several related records
What happens during failures?	Which reads and writes remain available and under what guarantees

Choose guarantees for the operation, not from the database label alone

Outline: NoSQL II

01	**Documents and Queries**	read fields, nesting and paths
02	**Finding Documents**	choose documents and output fields
03	**Summarizing Documents**	transform documents through ordered stages
04	**Updating Documents**	select write targets and verify changes

Collection and Document

Relational terms

Table	Row	Column
Product	one product	price

Document terms

Collection	Document	Field
inventory	one item	qty

A collection holds documents; each document has named fields

Fields, Objects and Arrays

One inventory document

```
{
  item: "paper",
  qty: 100,
  size: {h: 8.5, w: 11, uom: "in"},
  tags: ["red", "blank", "plain"],
  instock: [
    {loc: "A", qty: 60},
    {loc: "B", qty: 15}
  ]
}
```

Levels inside it

Field	Value
qty	number
size	nested object
tags	array of strings
instock	array of objects

A field can hold one value, a nested object or an array

Finding Documents

One inventory record

```
{item:"paper", qty:100, status:"D"}
```

Two choices

Choice	Purpose
Filter: status D	Keep matching documents
Fields: item and qty	Choose what the result shows

Choose documents and output fields before summarizing them

Documents and Nested Entries

One item document

```
item: "paper"
instock: [
  {loc:"A", qty:60},
  {loc:"B", qty:15}
]
```

Choose the relevant level

Question	Unit to inspect
Does one location satisfy both conditions?	The same stock entry
How much stock is recorded by location?	Each location entry, not each item document

Keep related conditions and measures at the same record level

A Pipeline Changes Its Input

Input documents

```
{state:"CA", pop:10}
{state:"CA", pop:20}
{state:"NY", pop:40}
```

Stages and outputs

Stage	Output
Keep pop ≥ 20	CA:20 · NY:40
Group by state; sum pop	CA total:20 · NY total:40

Each stage passes its output documents to the next stage

Filter, Group and Measure

Decision	Question	Example from the zip records
Filter	Which documents are included?	Keep state CA
Group	What does one output represent?	One city
Measure	What number do we compute?	SUM(pop), not the number of zip records

LOS ANGELES: 57,110 + 51,223 = 108,333 people across two zip records.

Choose the filter, grouping key and measure independently

Safe Document Updates

Step	Example
Preview the selected documents	The three inventory items with status A
Change the intended fields	Set those statuses to Review
Verify affected and unaffected records	Three changed; paper and planner stay D

Atomicity for one document does not automatically make a multi-document change atomic.

Preview the targets, apply the change, and verify its scope

Takeaway: Keep the Meaning Through Each Step

- Define what a record measures and which outcomes were observed
- Check what preparation changes, estimates or removes
- Choose a representation and a result that answer the question